

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ401			
Course Title	ADVANCED TOPOLOGY			
Type of Course	Elective			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial	Practicum	Total Hours
		per week	per week	
	4	4	0	60
Pre-requisites	1. Topology I			
Course Summary	The advanced topology course extends Topology I by introducing further concepts and tools. It starts with the product topology and explores its properties. Embeddings, including the Tychonoff embedding theorem, are discussed. Urysohn's Lemma from the previous course is used to prove the Urysohn Metrisation Theorem. Nets and filters are introduced to address sequence limitations. Various forms of compactness and compactifications are examined, with a focus on their relation to completeness in metric spaces. The course concludes with important results such as the Baire category theorems.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Learn basic structures and constructions in Topology	U	F	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Analyse and apply the concepts of Nets, Filters, and Convergence in the context of Topological Spaces	An	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	To develop the student's ability to handle abstract ideas of mathematics and mathematical proofs	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book	Introduction to General Topology, 2nd Edition, K. D. Joshi, New Age International Publishers, 1983.			
Module	Unit	Content	Hrs (48+12)	External Marks (70)
I	Chapter 8: Products & Coproducts		10	
	1	Cartesian Products of Families of Sets – 8.1		
	2	The Product Topology – 8.2		
	3	Productive Properties – Separation Axioms 8.3		
	4	Productive Properties – Connectedness – 8.3		
	5	Countably Productive Properties – Metrisability–8.4		
	6	Countably Productive Properties – Countability–8.4		
	7	The Case of Separability – 8.4		
II	Chapter 9: Embedding & Metrisation		10	
	8	Evaluation Functions into Products – 9.1		
	9	Embedding Lemma – 9.2		
	10	Tychonoff Embedding – 9.2		
	11	The Urysohn Metrisation Theorem – 9.3		
III	Chapter 10: Nets & Filters		12	
	12	Definition & Convergence of Nets – 10.1		
	13	Topology & Convergence of Nets – 10.2		
	14	Nets & Compactness – 10.2		
	15	Filters & Their Convergence – 10.3		
	16	Topology & Filters – 10.3		
	17	Ultrafilters and Compactness – 10.4		
IV	Chap 11,12: Compactness & Complete Metric Spaces		16	

	18	Variations of Compactness – 11.1		
	19	The Alexander Sub-base Theorem – 11.2		
	20	Local Compactness – 11.3		
	21	Compactifications – 11.4 (Wallman Compactification 11.15 to 11.20 may be relegated to Practicum)		
	22	Complete Metrics – 12.1		
	23	Consequences of Completeness – 12.2		
	24	Completions of a Metric – 12.4		
V			12	
	1	Wallman Compactification: 11.15 to 11.20		
	2	12.3: Some Applications (of Completeness)		
	3	Chapter 13: Category Theory		
	4	Chapter 14: Uniform Spaces		
	5	Chapter 15 Section 2: Paracompactness		
	6	Chapter 15 Section 3: Use of Ordinal Numbers		
	7	Nagata-Smirnov Metrisation Theorem		
References 1. Topology, J. R. Munkres, Prentice Hall of India, 2000. 2. General Topology, S. Willard, Addison Wesley Pub. Company, 1976. 3. General Topology, J. L. Kelley, D. van Nostrand, 1955. 4. Introduction to Topology and Modern Analysis; G. F. Simmons, McGraw-Hill, 1963. 5. Topology, James Dugundji, Prentice Hall of India, 1975.				

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	3	1	3	1	2	1	3	0	1
CO 2	3	3	2	1	3	1	3	1	3	0	1
CO 3	3	3	3	3	2	1	2	1	2	0	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓